

Credit Risk Analysis

End-to-End Data Analytics Project



2026

Tools Used:-

Python | SQL | Power BI | Matplotlib | Seaborn

*~ Prepared By
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Project Overview

The objective of this project is to analyze historical loan application data to identify factors influencing loan defaults. The project uses Python for data cleaning and exploratory data analysis (EDA), SQL for business-oriented analysis, and Power BI for interactive dashboard development. The insights help financial institutions identify high-risk borrowers and support better lending decisions.

Data Cleaning & Feature Engineering

Missing Value Treatment

The dataset was examined for missing values to ensure completeness before analysis. Missing records were removed to maintain data quality and prevent inaccurate statistical results.

Outlier Treatment

Extreme and unrealistic values were identified in the Age and Employment Length variables using exploratory analysis. For example, borrower ages and employment durations that were not practically possible were removed from the dataset. Eliminating these records improved the reliability of the analysis by ensuring that summary statistics and visualizations were not distorted by erroneous data.

Duplicate Record Check

The dataset was checked for duplicate records to ensure that each loan application represented a unique observation. No duplicate entries were retained in the final dataset.

Feature Engineering:-

Age Group

The continuous age variable was categorized into meaningful age groups (18–25, 26–35, 36–45, 46–60, and 60+) to simplify borrower segmentation and facilitate comparative analysis across different age categories.

Income Category

Annual income was divided into four equally sized categories using quartile-based binning (Low, Medium, High, and Very High). This transformation enabled a balanced comparison of default rates across different income levels.

Exploratory Data Analysis (EDA)

Correlation Heatmap

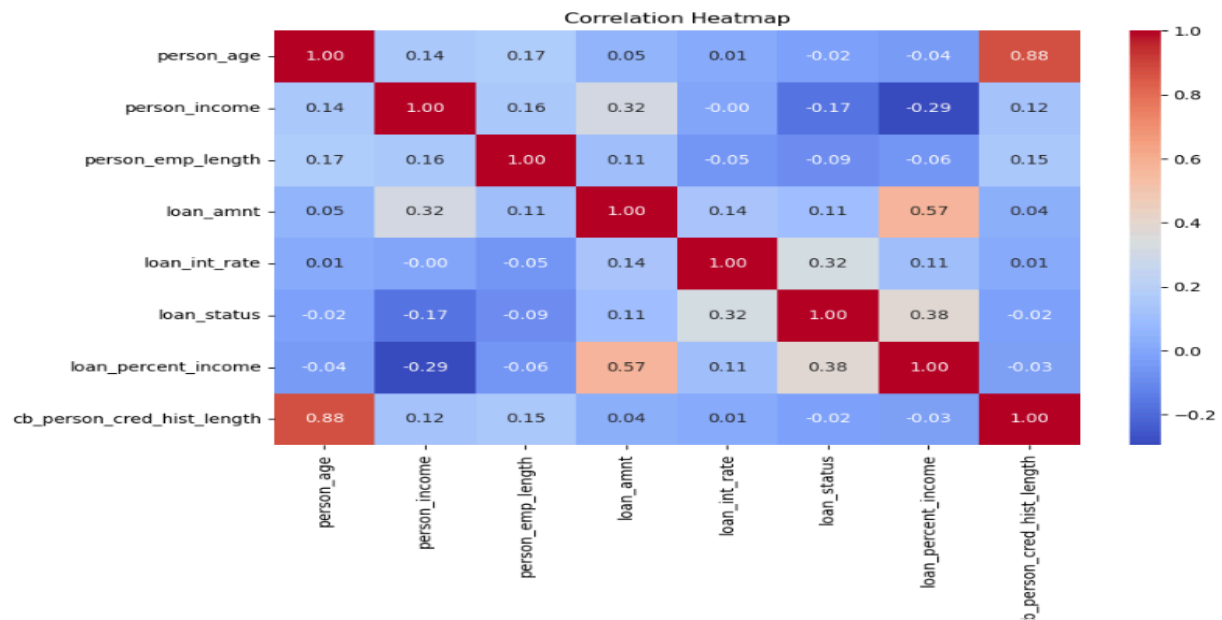
```
import matplotlib.pyplot as plt
import seaborn as sns

# Select only numerical columns
numeric_data = clean_data .select_dtypes(include="number")

# Calculate correlation
correlation_matrix = numeric_data.corr()

# Create the heatmap
plt.figure(figsize=(10,6))
sns.heatmap(correlation_matrix,
            annot=True,
            cmap="coolwarm",
            fmt=".2f")

plt.title("Correlation Heatmap")
plt.show()
```



Purpose:- To examine the relationships between numerical variables and identify patterns that may influence loan default behavior.

Key Findings

- Age and credit history length exhibit a strong positive correlation.
- Higher loan amounts are associated with higher loan-to-income ratios.
- Borrowers with higher interest rates tend to show a higher likelihood of default.

Scatter Plot (Income vs Loan Amount)

```
import matplotlib.pyplot as plt
import seaborn as sns

plt.figure(figsize=(10,6))

sns.scatterplot(
    data=clean_data,
    x="person_income",
    y="loan_amnt",
    hue="loan_status"
)

plt.title("Income vs Loan Amount by Loan Status")
plt.xlabel("Annual Income")
plt.ylabel("Loan Amount")
plt.show()
```



Purpose

To analyze the relationship between borrower income and loan amount while distinguishing borrowers based on loan repayment status.

Key Findings

- Borrowers with higher incomes generally qualify for larger loans.
- Loan defaults are observed across different income levels but appear more concentrated among lower-income borrowers.

Income Distribution by Loan Status (Box Plot)

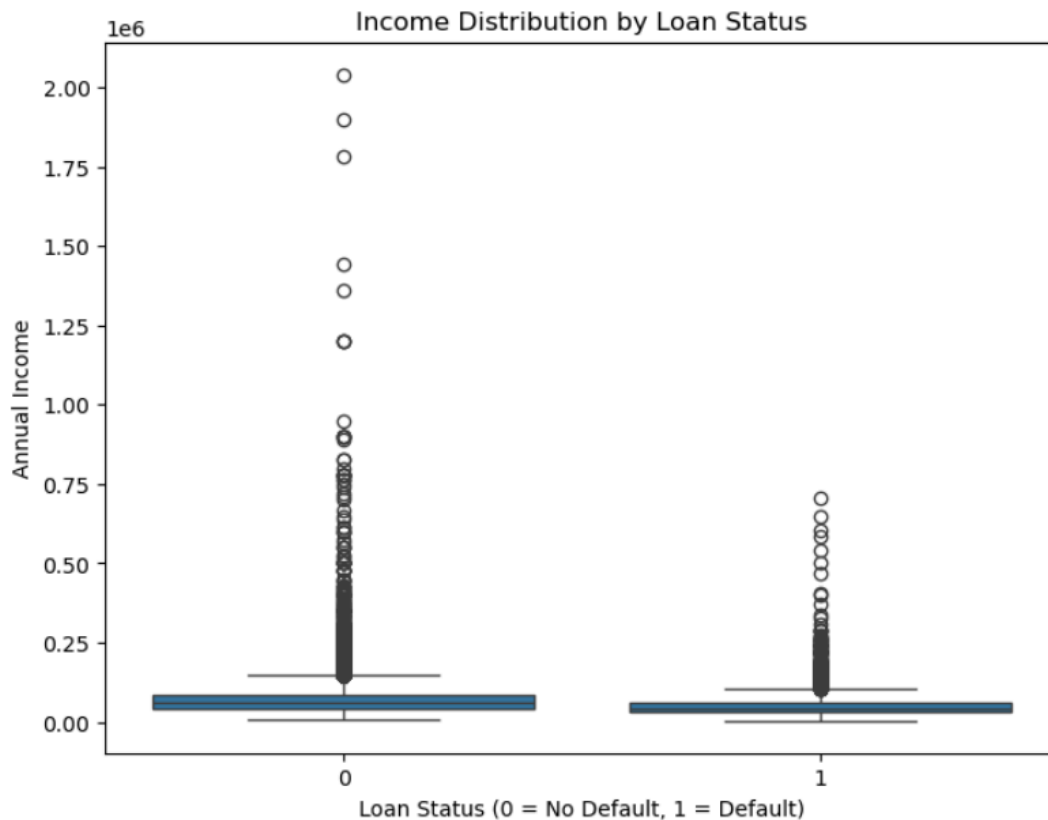
```
import matplotlib.pyplot as plt
import seaborn as sns

plt.figure(figsize=(8,6))

sns.boxplot(
    data=clean_data,
    x="loan_status",
    y="person_income"
)

plt.title("Income Distribution by Loan Status")
plt.xlabel("Loan Status (0 = No Default, 1 = Default)")
plt.ylabel("Annual Income")

plt.show()
```



Purpose

To compare the income distribution of borrowers who repaid their loans with those who defaulted.

Key Findings

- Borrowers who defaulted generally exhibit lower income levels than those who successfully repaid their loans.
- The presence of income outliers indicates considerable variation within borrower groups.

Distribution of Loan Amount (Histogram)

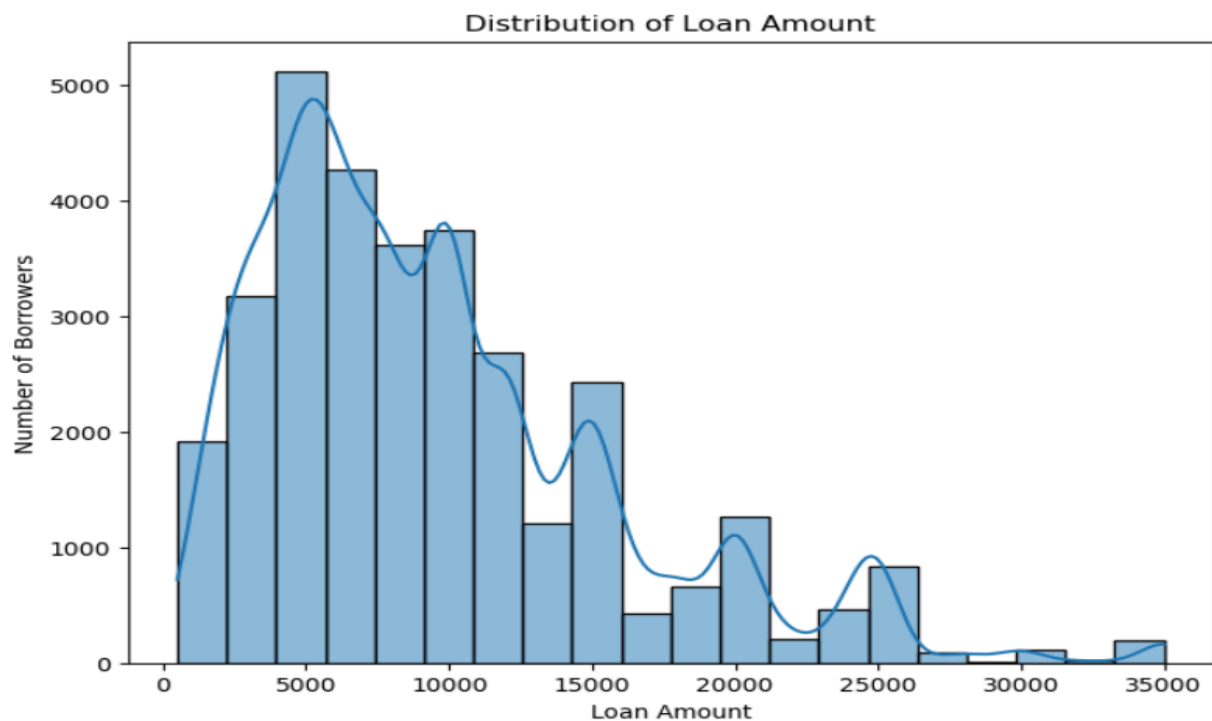
```
import matplotlib.pyplot as plt
import seaborn as sns

plt.figure(figsize=(8,6))

sns.histplot(
    data=clean_data,
    x="loan_amnt",
    bins=20,
    kde=True
)

plt.title("Distribution of Loan Amount")
plt.xlabel("Loan Amount")
plt.ylabel("Number of Borrowers")

plt.show()
```



Purpose

To understand how loan amounts are distributed across all borrowers.

Key Findings

- Most borrowers applied for relatively small to medium-sized loans.
- Large loan amounts are comparatively less common, resulting in a positively skewed distribution.

SQL Business Analysis

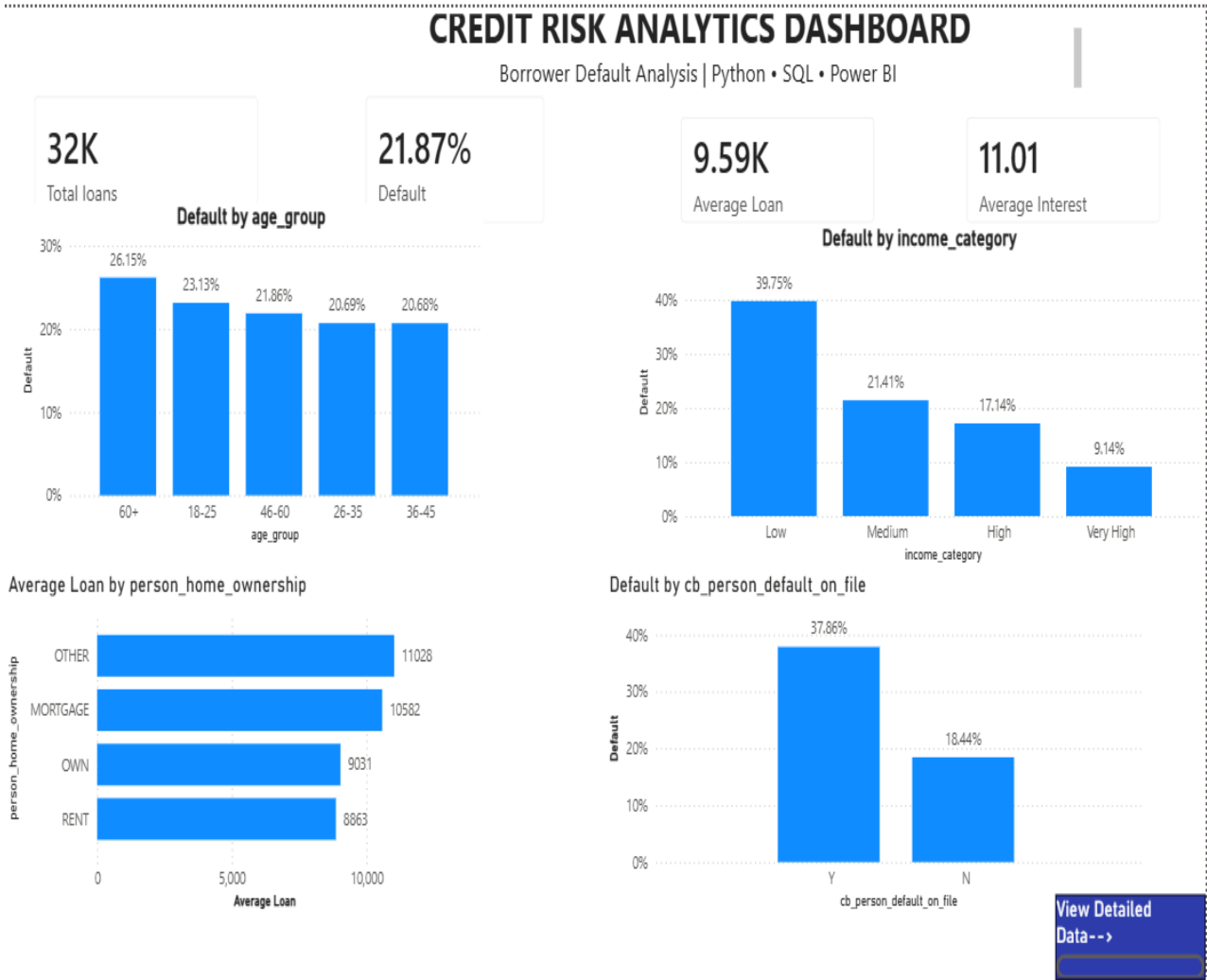
Business Question	Key Insight
What is the overall loan default rate?	Approximately 21.87% of borrowers defaulted on their loans, indicating that nearly one in five loan applications resulted in default.
Does age affect the likelihood of default?	Younger borrowers exhibited higher default rates compared to older age groups, suggesting that age may influence repayment behavior.
Does income affect the likelihood of default?	Borrowers with lower annual incomes showed higher default rates, while higher-income borrowers demonstrated better repayment performance.
Which loan purposes have the highest default rates?	Debt Consolidation loans recorded the highest default rate (28.68%), followed by Medical (26.76%) and Home Improvement (26.15%) loans. In contrast, Venture loans had the lowest default rate (14.86%), indicating that loan purpose has a significant influence on repayment.
How does loan grade affect default risk?	Borrowers assigned lower credit grades (such as D, E, F, and G) experienced significantly higher default rates than those with better credit grades.
Which home ownership type receives the highest average loan amount?	Borrowers with Other home ownership status received the highest average loan amount (11,028.07), followed by Mortgage holders (10,581.89). Borrowers who own (9,031.10) or Rent (8,863.43) their homes received comparatively smaller average loan amounts.
Does previous default history affect future defaults?	Borrowers with a previous default history had a substantially higher likelihood of defaulting again, making past repayment

behavior a strong indicator of future credit risk.

Which borrower segments are the highest risk?

Young borrowers with lower incomes and rented homes represented the highest-risk borrower segment in the dataset.

Power BI Dashboard



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Default Rate by Age Group, Income Category and Home Ownership

age_group	High	Low	Medium	Very High	Total
18-25	16.91%	40.59%	21.83%	8.09%	23.13%
MORTGAGE	13.08%	24.57%	13.84%	6.86%	12.85%
OTHER	33.33%	60.00%	40.00%	0.00%	32.81%
OWN	2.18%	16.70%	0.95%	1.17%	7.66%
RENT	22.52%	48.94%	29.22%	11.84%	32.91%
26-35	17.11%	39.44%	20.76%	9.41%	20.69%
MORTGAGE	14.15%	20.49%	15.51%	8.14%	12.48%
OTHER	18.18%	71.43%	20.00%	0.00%	27.59%
OWN	4.21%	17.00%	1.54%	1.30%	7.13%
RENT	22.29%	48.68%	26.67%	13.74%	30.24%
36-45	17.56%	35.05%	21.50%	11.36%	20.68%
MORTGAGE	14.29%	16.24%	13.87%	8.86%	12.03%
OTHER	0.00%	50.00%		50.00%	40.00%
OWN	0.00%	17.00%	1.64%	1.96%	7.57%
RENT	23.10%	44.52%	29.47%	17.70%	30.54%
46-60	17.69%	44.04%	23.74%	10.75%	21.86%
MORTGAGE	19.40%	29.17%	7.55%	8.11%	12.94%
OTHER	0.00%		0.00%	0.00%	0.00%
OWN	0.00%	38.46%	0.00%	0.00%	10.87%
RENT	18.31%	50.00%	40.85%	17.46%	32.13%
60+	57.14%	27.27%	21.05%	9.52%	26.15%
MORTGAGE	66.67%		0.00%	12.50%	18.18%
RENT	54.55%	27.27%	25.00%	0.00%	30.23%
Total	17.14%	39.75%	21.41%	9.14%	21.87%

Business Recommendations

- **Strengthen credit evaluation for applicants with a history of previous defaults.**
- **Apply additional risk assessment for lower loan grades due to their higher default rates.**
- **Monitor younger and lower-income borrower segments more closely during loan approval.**
- **Consider income-to-loan ratio when evaluating repayment capacity.**
- **Use borrower segmentation to support data-driven lending decisions.**

Conclusion:-

This project demonstrates an end-to-end data analytics workflow by integrating Python, SQL, and Power BI to analyze credit risk. Through data cleaning, exploratory analysis, business-oriented SQL queries, and interactive dashboards, the project identifies key borrower characteristics associated with loan defaults. The insights obtained can assist financial institutions in improving risk assessment and making informed lending decisions.

